



L5a: Single Asset Geometric Brownian Motion and NPV

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

By the end of this lecture, you will be able to:

- **Interpret out-of-sample predictions:** explain how historical mean growth and volatility determine the distribution of later prices.
- **Calculate an NPV target probability:** connect the discounted trade payoff to the probability of exceeding a target at the sale date.
- **Compare predictions with observations:** use the examples to assess price predictions and examine whether updating improves forecasts.

Lectures and Examples

Example: Out-of-Sample Single-Asset GBM

How well do historical estimates describe later prices? Simulate 2025 prices, construct prediction bands, and compare observed coverage across assets.

Example: GBM Net Present Value Trade Rule

How likely is a trade to exceed a target scaled NPV? Calculate the probability, check it at the median, and examine how it changes with the target.

Example: Updating GBM Parameters with EMA

Can recent observations improve the forecasts? Update mean growth and volatility, then compare the resulting forecasts with the frozen model.

Company Profile: Renaissance Technologies

Renaissance Technologies (RenTec) is a quantitative investment management firm founded in 1982. Its investment work uses mathematical and statistical methods, with a data-driven research approach across its funds.

Its researchers include people trained in mathematics, physics, computer science, and related fields.

Connection to today: State the assumptions, record a forecast, and evaluate it against later observations. Our GBM example is a teaching model, not a description of RenTec's proprietary methods.

Sources: [RenTec overview](#); [firm description](#).

Concept Review: Single Asset GBM Model

Recall that geometric Brownian motion (GBM) describes share price changes through drift and random fluctuations, both proportional to the current price:

$$\frac{dS(t)}{S(t)} = \underbrace{\mu dt}_{\text{drift}} + \underbrace{\sigma dW(t)}_{\text{random fluctuations}}.$$

Here, $S(t)$ is the share price (dollars per share), with initial price $S(0) = S_0 > 0$, and $dW(t)$ is a Wiener-process increment.

With time measured in years, the constant parameters are:

- **Drift** $\mu \in \mathbb{R}$: the expected proportional price change per unit time (year^{-1}).
- **Volatility** $\sigma > 0$: sets the size of the random fluctuations ($\text{year}^{-1/2}$).

Concept Review: Growth and Volatility

Over an observation interval $\Delta t > 0$ years, the growth rate is:

$$g_j = \frac{1}{\Delta t} \ln \left(\frac{S_{t_j}}{S_{t_{j-1}}} \right), \quad t_j = j\Delta t.$$

Under GBM, its mean and standard deviation are:

$$\underbrace{\mu_g = \mathbb{E}[g_j] = \mu - \frac{\sigma^2}{2}}_{\text{mean growth rate}}, \quad \underbrace{\sqrt{\text{Var}(g_j)} = \frac{\sigma}{\sqrt{\Delta t}}}_{\text{growth-rate standard deviation}}.$$

Drift μ and mean growth rate μ_g both have units year^{-1} , but differ by $\sigma^2/2$.

From the sample growth-rate standard deviation σ_g , we estimate volatility as:

$$\hat{\sigma} = \sigma_g \sqrt{\Delta t}.$$

Concept Review: The One-Step Transition

For a grid of N steps, $t_j = j\Delta t$, the exact transition from L4b is:

$$S_{t_j} = S_{t_{j-1}} \exp\left[\mu_g \Delta t + \sigma \sqrt{\Delta t} Z_j\right], \quad j = 1, \dots, N.$$

Here, $Z_1, \dots, Z_N$ are independent standard normal draws, one per step.

For a single horizon $T > 0$, we can sample the terminal price directly:

$$S_T = S_0 \exp\left[\mu_g T + \sigma \sqrt{T} Z\right], \quad Z \sim \mathcal{N}(0, 1).$$

Repeated one-step draws generate a price path. The fixed-horizon form samples the price at the selected time T .

Example: Out-of-Sample Price Predictions

Example: Test predictions against later observations

How well does a model fitted to historical prices describe prices observed afterward?

- Use the mean growth rate and volatility estimated from 2014–2024 data.
- Simulate prices for 2025 and compare prediction bands with observed prices.
- Measure the fraction of dates with observed prices inside each band, first for one firm and then across the dataset.

Next question: how can this price distribution tell us the probability of exceeding a target return at a scheduled sale?

GBM Trade Rule

Consider a scheduled trade: buy $n_0 > 0$ shares at S_0 at time 0 and sell at S_T at time $T = N\Delta t$. Assume no dividends or trading costs.

Let g_y be the constant, continuously compounded benchmark growth rate (year^{-1}) corresponding to yield y . The NPV is given by:

$$\text{NPV}(g_y, T) = \underbrace{-n_0 S_0}_{\text{purchase today}} + \underbrace{n_0 S_T e^{-g_y T}}_{\text{discounted sale}}.$$

Dividing by the initial investment gives the **scaled NPV**, our discounted fractional return:

$$\rho_T = \frac{\text{NPV}(g_y, T)}{n_0 S_0} = \left(\frac{S_T}{S_0} \right) e^{-g_y T} - 1.$$

We now use GBM to describe the uncertain sale price S_T .

Substitute the GBM Solution

Using mean growth rate $\mu_g = \mu - \sigma^2/2$, substitute the GBM price into the scaled NPV:

$$\rho_T = \exp\left[(\mu_g - g_y)T + \sigma\sqrt{T} Z\right] - 1, \quad Z \sim \mathcal{N}(0, 1).$$

Discounting subtracts the benchmark growth rate g_y from the mean growth rate μ_g , while leaving the random shock unchanged. The quantity $1 + \rho_T$ is lognormal, so:

$$\ln(1 + \rho_T) \sim \mathcal{N}\left((\mu_g - g_y)T, \sigma^2 T\right).$$

The scaled NPV is a lognormal variable shifted down by one, with $\rho_T > -1$.

The benchmark affects the return through $g_y T$. When $|g_y|T \ll 1$, discounting is small and $\rho_T \approx S_T/S_0 - 1$. We use the exact discounted expression to calculate the target probability.

Terminal Target Probability

Choose a target scaled NPV $\rho_\star > -1$. Adding one and taking the logarithm preserves the strict target condition:

$$\rho_T > \rho_\star \iff \ln(1 + \rho_T) > \ln(1 + \rho_\star).$$

Substituting the GBM expression gives:

$$(\mu_g - g_y)T + \sigma\sqrt{T} Z > \ln(1 + \rho_\star).$$

Subtracting $(\mu_g - g_y)T$ and dividing by $\sigma\sqrt{T} > 0$ gives a threshold for the standard normal random variable Z :

$$Z > z_\star, \quad \boxed{z_\star = \frac{\ln(1 + \rho_\star) - (\mu_g - g_y)T}{\sigma\sqrt{T}}}.$$

Thus $Z > z_\star$ means the trade exceeds its target at the scheduled sale. Selling when a boundary is first reached requires a separate calculation.

The Closed-Form Target Probability

The standard normal cumulative distribution function $\Phi(z_\star)$ gives the probability that $Z \leq z_\star$. Subtracting from one gives the probability that $Z > z_\star$:

$$\mathbb{P}(\rho_T > \rho_\star) = 1 - \Phi(z_\star) = 1 - \Phi\left(\frac{\ln(1 + \rho_\star) - (\mu_g - g_y)T}{\sigma\sqrt{T}}\right).$$

- For a chosen target, we evaluate $z_\star$ from estimated mean growth and volatility, the holding period, and benchmark growth rate.
- The formula applies for $T > 0$, $\sigma > 0$, and $\rho_\star > -1$.
- A target of -1 or below is exceeded with probability one because the scaled NPV always exceeds -1 under GBM.

Example: Calculate and verify the NPV target probability

Change the Benchmark

The target scaled NPV corresponds to a required sale price:

$$K = S_0(1 + \rho_\star)e^{g_y T}.$$

With $\rho_\star = 0$, the trade succeeds if it beats the benchmark.

- $g_y = 0$: did the price increase?
- Positive g_y : did it increase enough over the holding period?
- Increasing g_y raises the sale threshold and lowers its probability.

The same prices can be assessed against different assumed growth benchmarks. Choose the benchmark before examining the outcome.

Example: GBM Net Present Value Trade Rule

Evaluate an NPV-based GBM trade rule

How likely is a stock trade to exceed our target scaled NPV at a scheduled sale?

We calculate this probability using the selected firm's mean growth and volatility, the holding period, and the benchmark growth rate.

We check the calculation at the median scaled NPV, which must be exceeded with probability one half, then vary the target to examine the probability curve.

Example: Updating the Parameter Estimates

Compare frozen and updated estimates

Can giving recent observations more weight improve our forecasts?

We retain the original purchase and update its target probability each day for a sale at the end of the chosen forward window. The example compares frozen parameters with exponentially weighted estimates of volatility alone or both mean growth and volatility.

We compare the forecasts on the same observed outcomes to examine whether the updates improve predictions or add noise.

Optional Advanced Material

The L4b examples develop two related questions:

- **Uncertainty in mean growth**: why can an estimate remain uncertain even with thousands of observations? Examine the observation period, sampling frequency, and their effects on target probabilities.
- **Monte Carlo versus the closed form**: how many paths are needed to estimate a probability accurately? Compare simulation error with the analytical result and introduce variance reduction techniques.

Examples: Mean-growth uncertainty; Monte Carlo target probabilities.

Summary

We connected the single-asset GBM model with discounted trade payoffs and observed outcomes.

- **Out-of-sample predictions:** we used historical estimates to describe the distribution of later prices.
- **NPV target probabilities:** we derived the chance of exceeding a target and connected it to the chosen benchmark growth rate.
- **Predictions and observations:** we checked price predictions and examined whether updated estimates improve target probabilities.

Next: L5b extends the price model to correlated assets and portfolio weights.